



(12) **United States Patent**
Striebich et al.

(10) **Patent No.:** **US 12,416,308 B2**
(45) **Date of Patent:** **Sep. 16, 2025**

(54) **COMPRESSOR WITH SHUTDOWN ASSEMBLY**

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

(21) Appl. No.: **18/394,474**

(22) Filed: **Dec. 22, 2023**

(65) **Prior Publication Data**

US 2024/0218881 A1 Jul. 4, 2024

(30) **Foreign Application Priority Data**

Dec. 28, 2022 (IN) 202221076341

(51) **Int. Cl.**

F04C 18/02 (2006.01)

F04C 27/00 (2006.01)

(Continued)

(52) **U.S. Cl.**

CPC **F04C 29/12** (2013.01); **F04C 18/0215** (2013.01); **F04C 18/0261** (2013.01); **F04C 27/005** (2013.01); **F04C 28/26** (2013.01)

(58) **Field of Classification Search**

CPC .. **F04C 29/12**; **F04C 18/0215**; **F04C 18/0261**;
F04C 27/005; **F04C 28/26**

See application file for complete search history.

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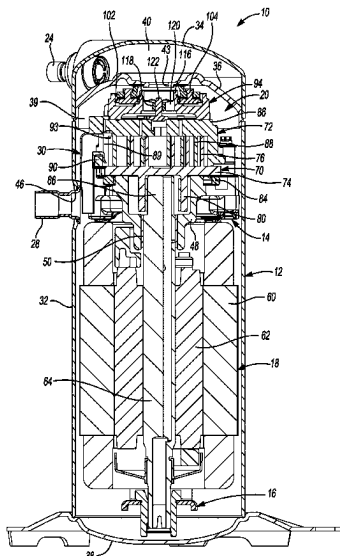
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(57) **ABSTRACT**

A compressor may include first and second scrolls, and a shutdown assembly. The first and second scrolls have first and second spiral wraps, respectively. The second scroll includes a discharge passage that receives compressed working fluid from a pocket defined by the first and second spiral wraps. The shutdown assembly may include a valve housing and a valve member. The valve housing may include a first aperture that receives working fluid from the discharge passage. The valve member may include a valve stem and a valve head. The valve stem may be movably received in a second aperture of the valve housing and has a smaller diameter than a diameter of the valve head. The valve member may be movable between a closed position in which the valve head restricts flow through the discharge passage and an open position in which the valve head allows flow through the discharge passage.

11 Claims, 8 Drawing Sheets



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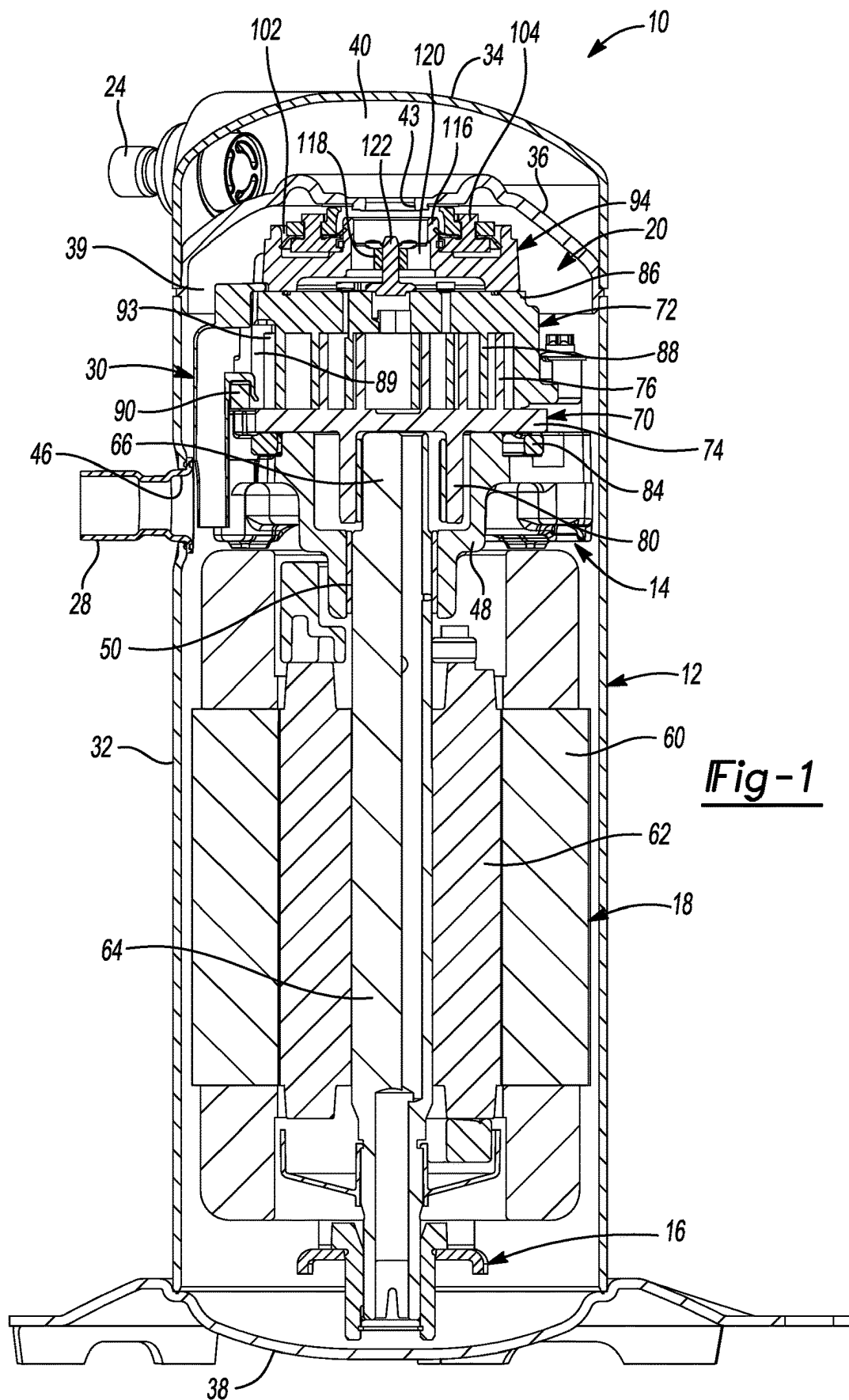
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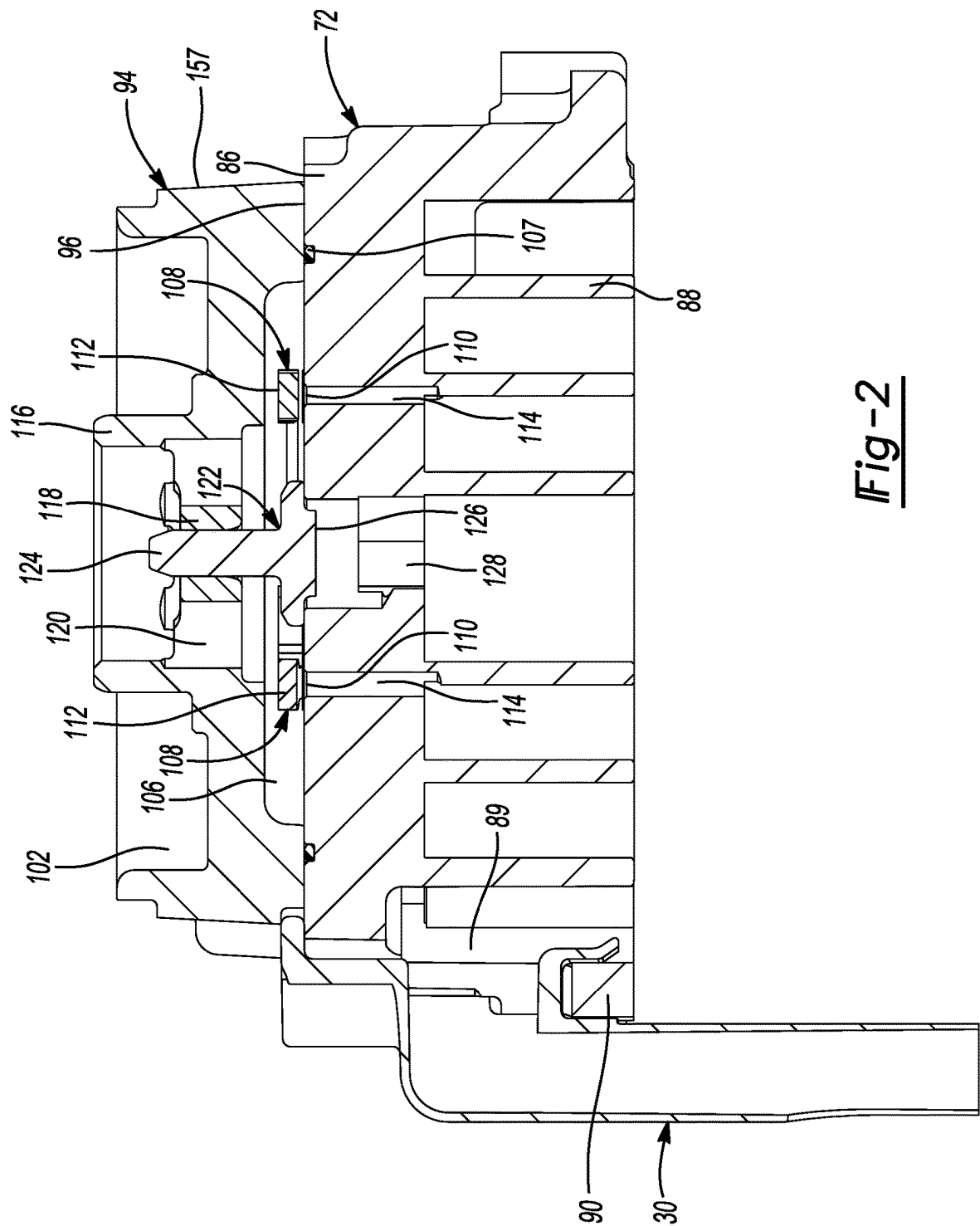


Fig-2

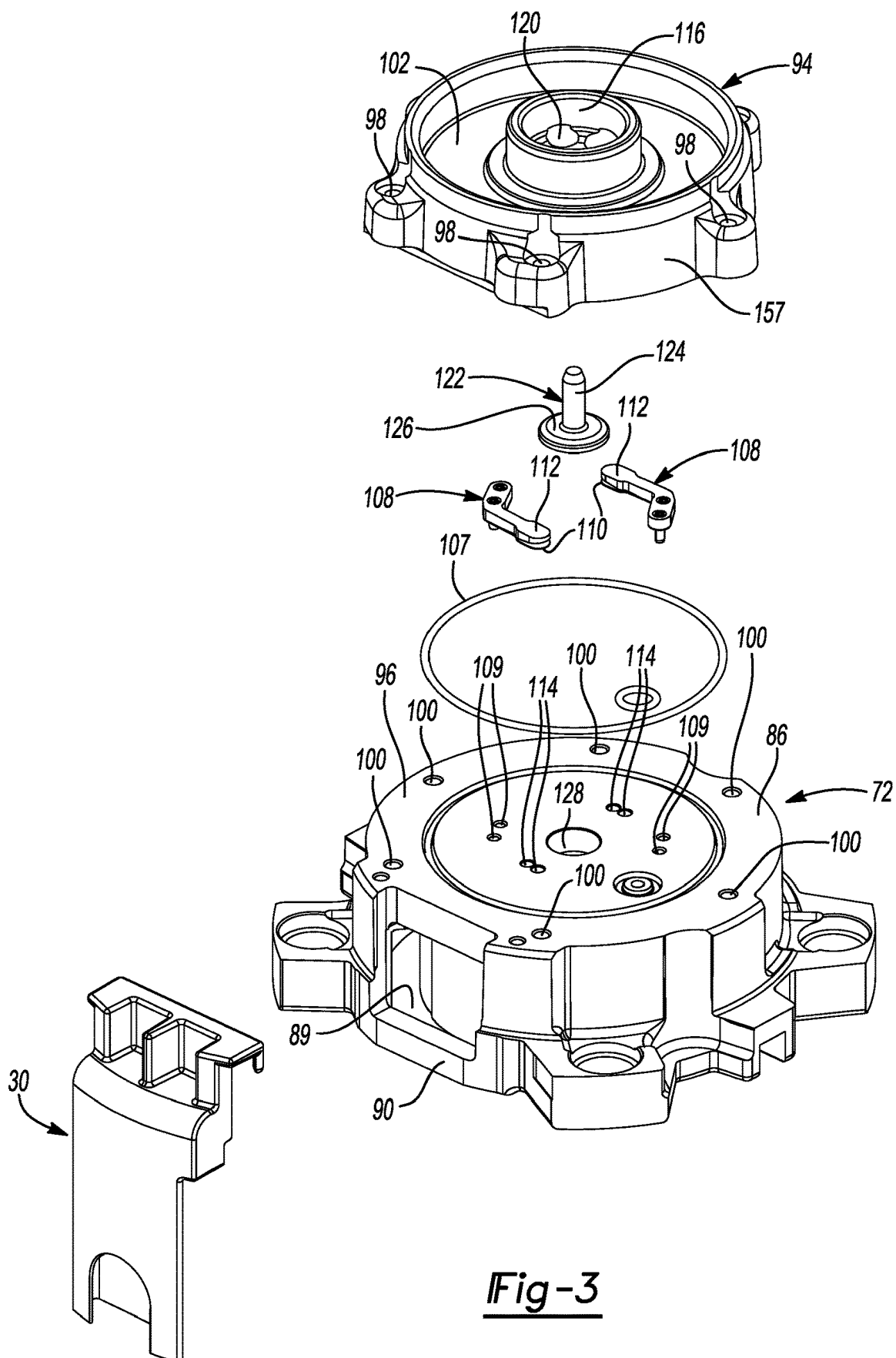


Fig-3

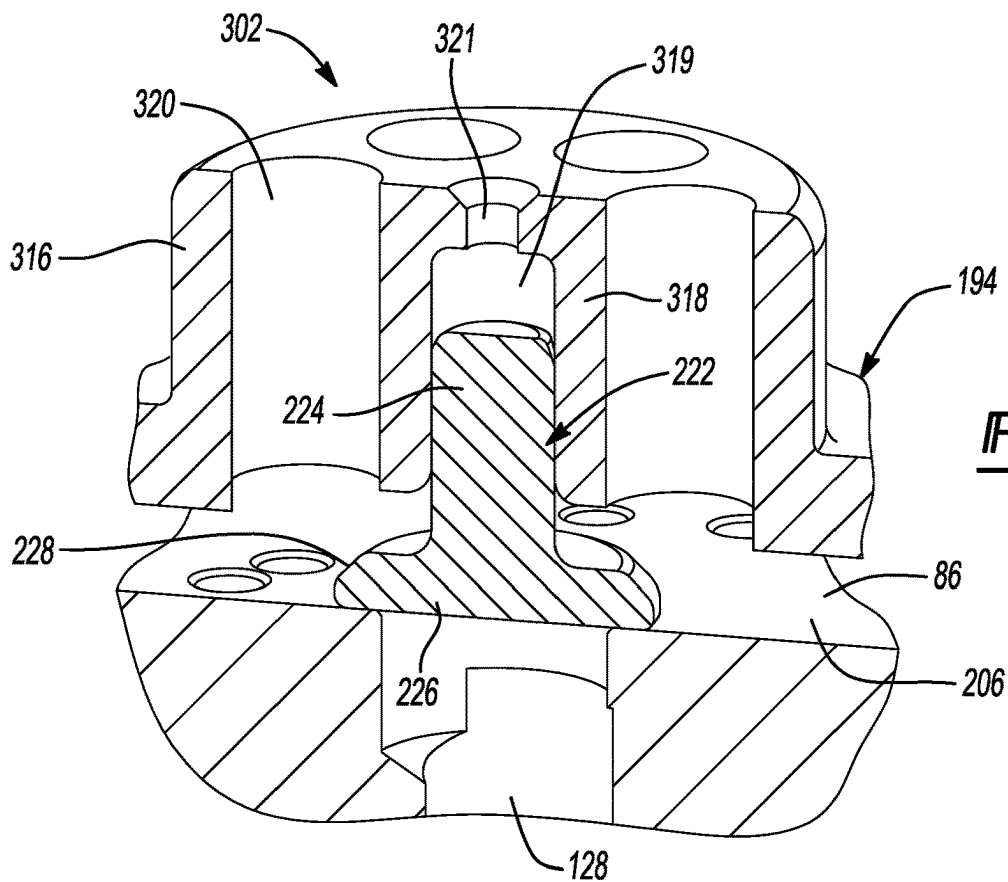


Fig-4

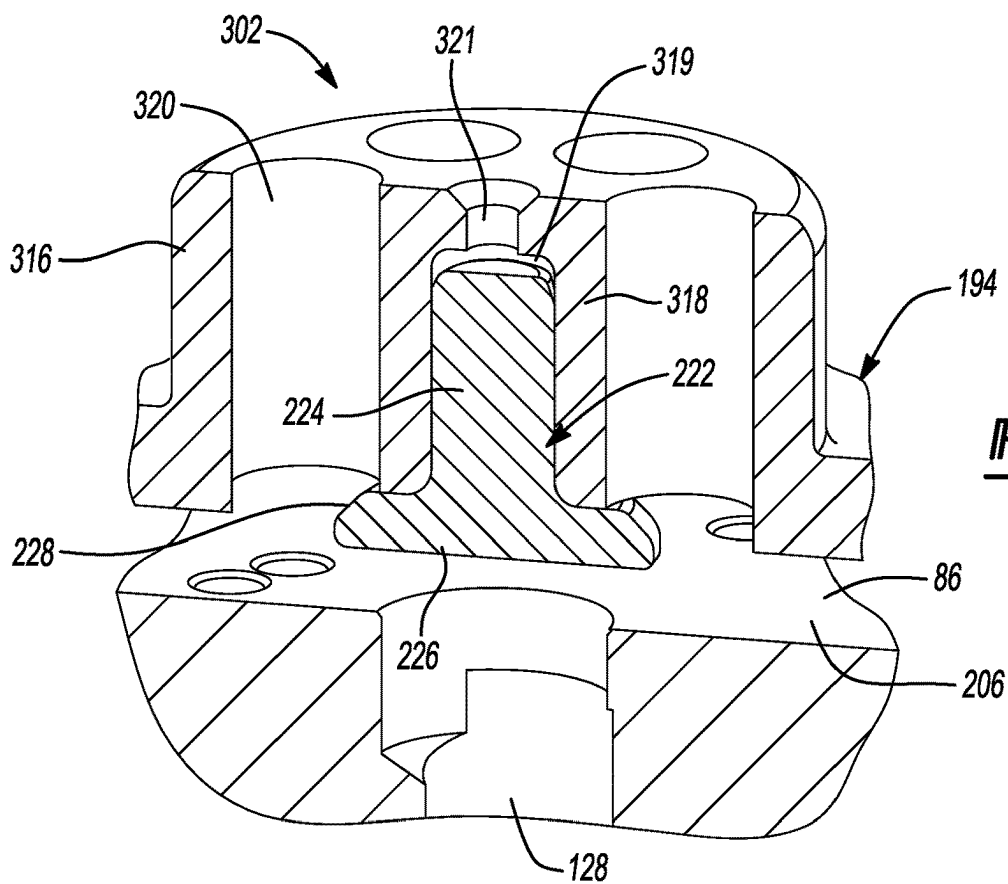
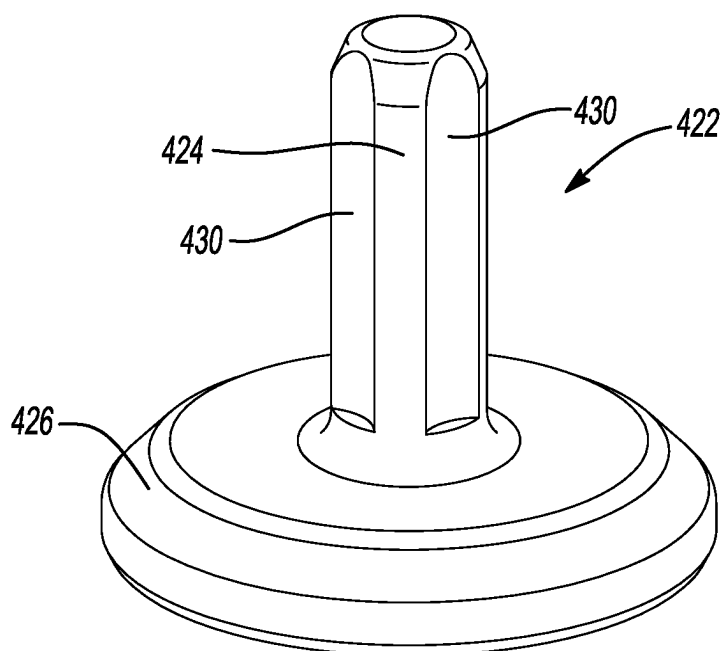
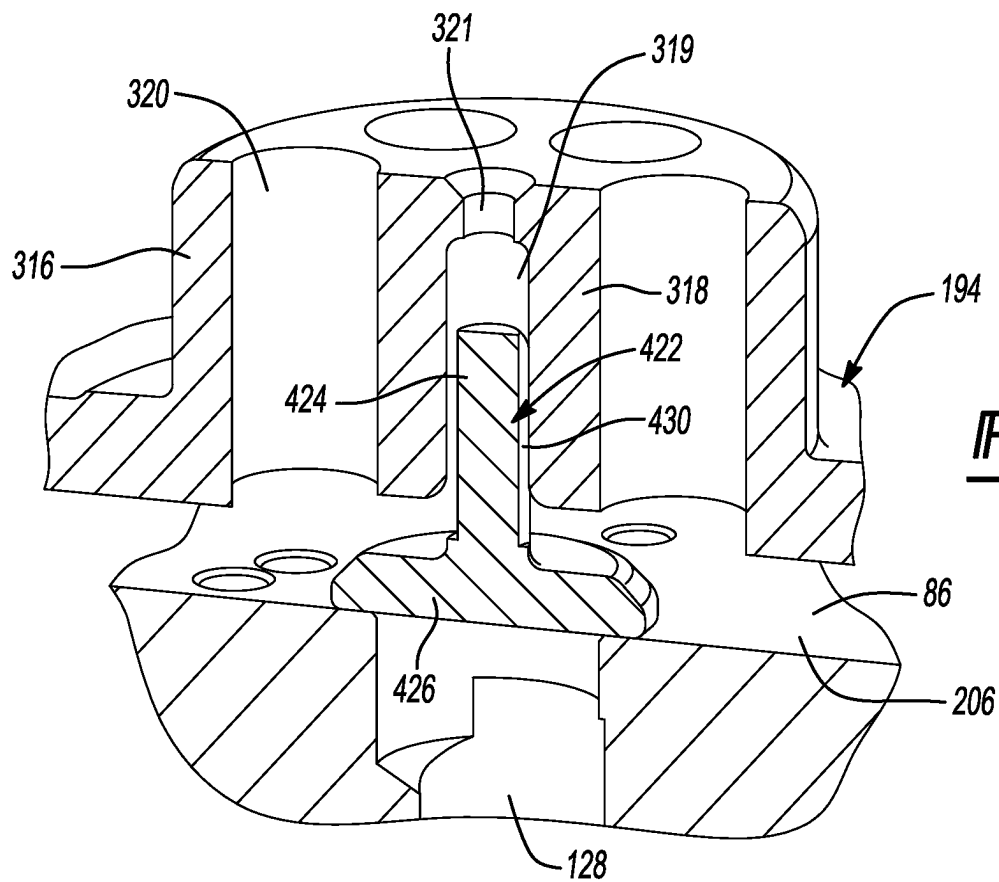
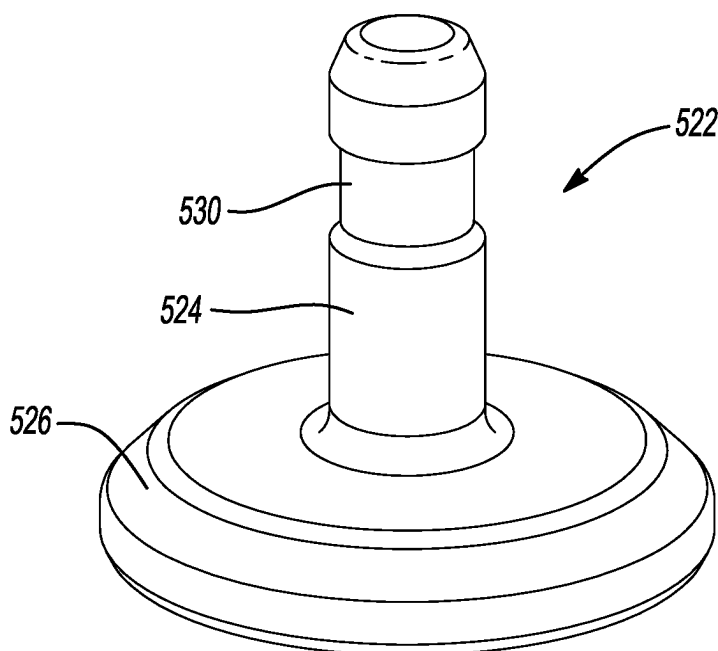
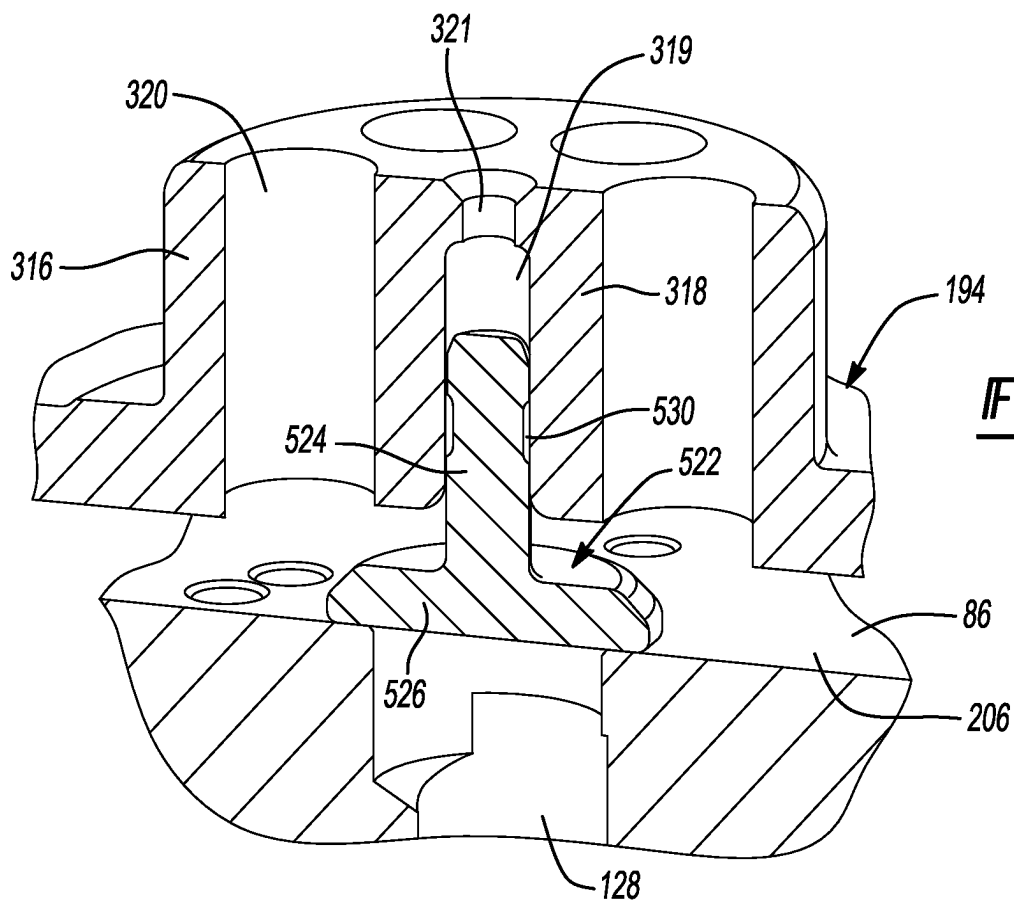
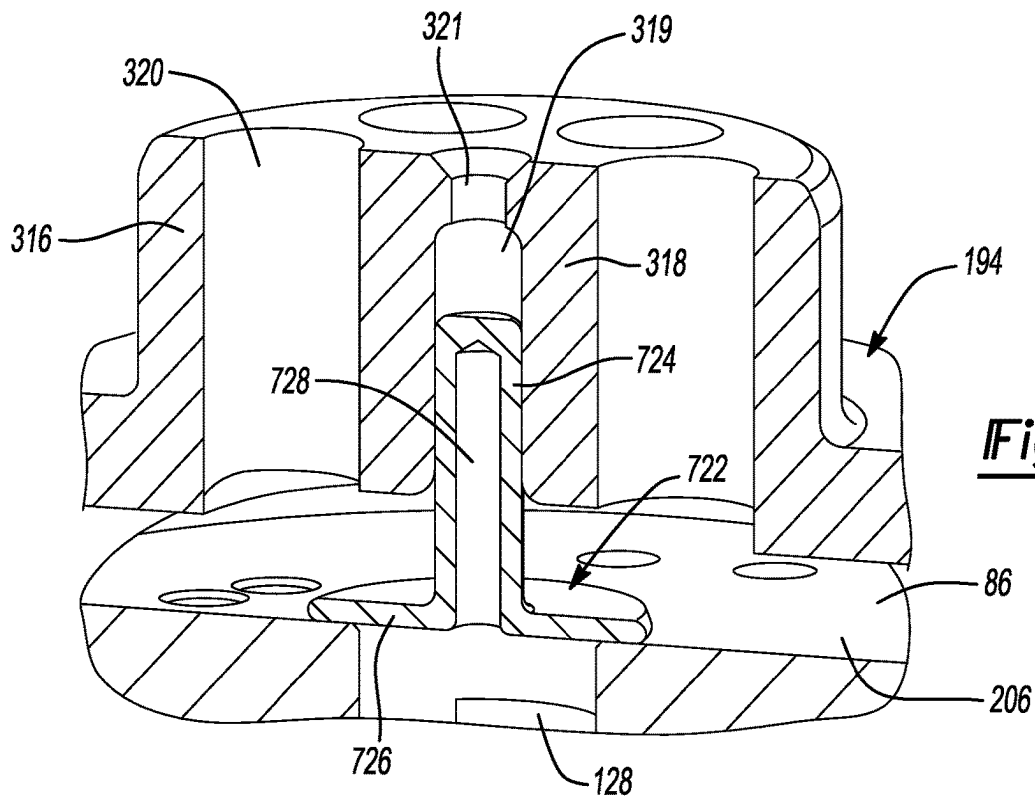
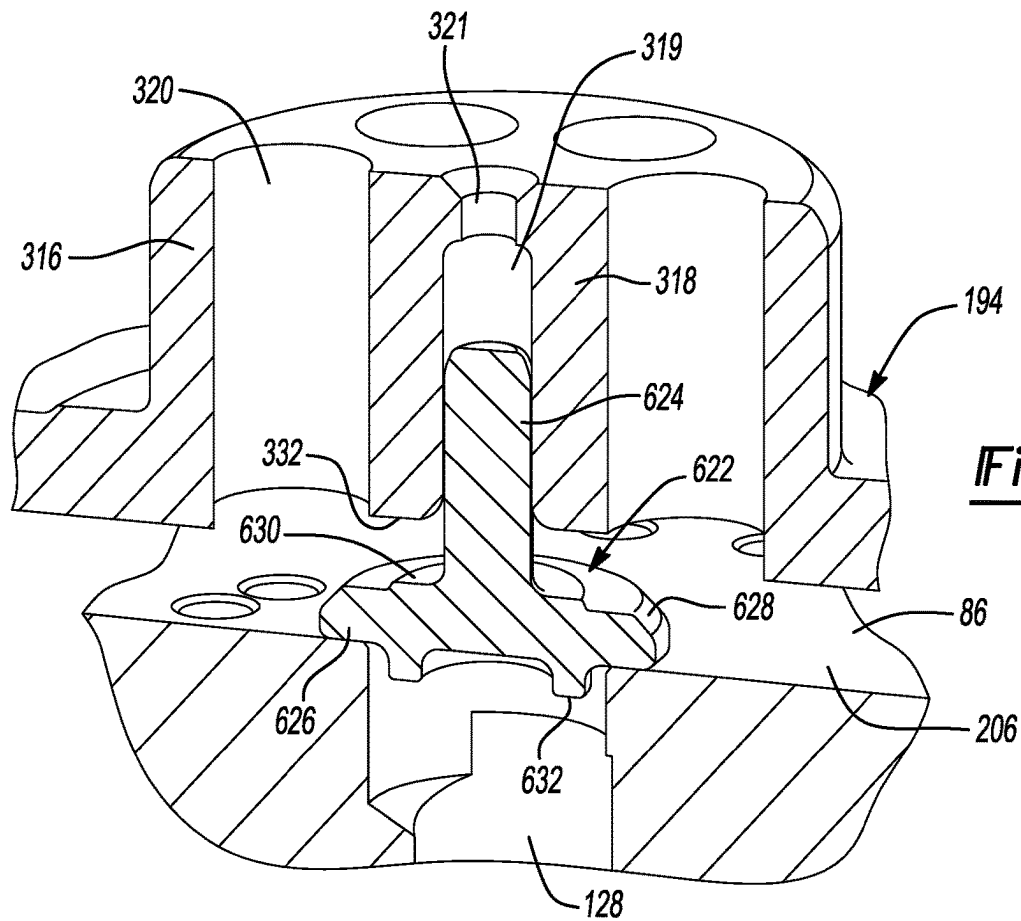


Fig-5







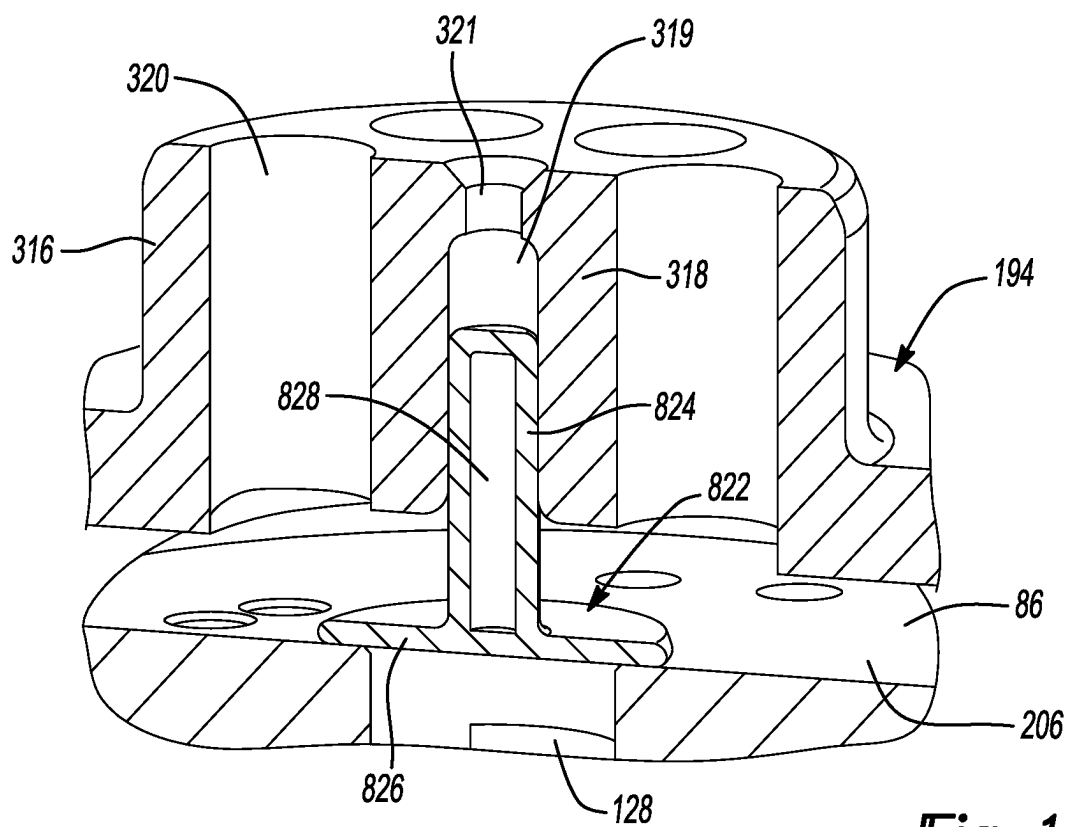


Fig-12

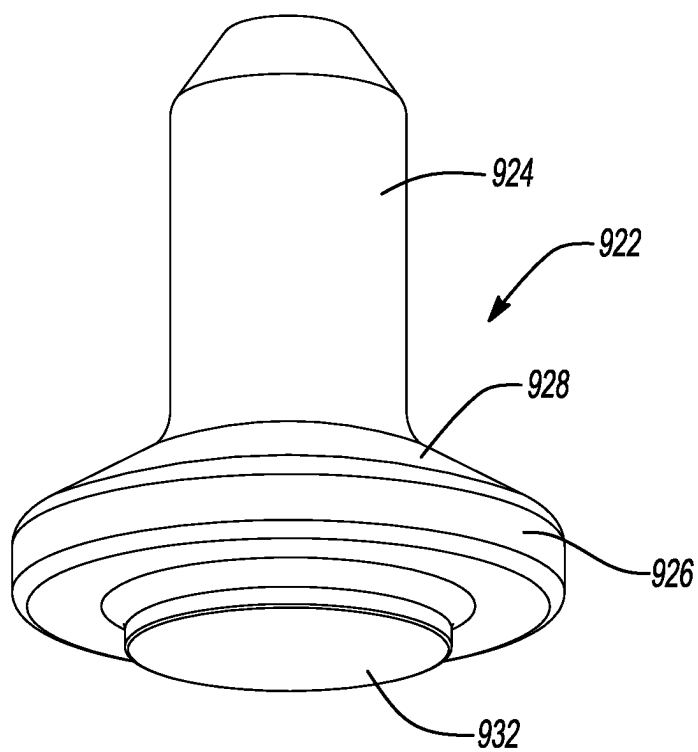


Fig-13

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**COMPRESSOR WITH SHUTDOWN
ASSEMBLY****CROSS-REFERENCE TO RELATED
APPLICATIONS**

This application claims the benefit and priority of Indian Patent Application No. 202221076341, filed Dec. 28, 2022. The entire disclosure of the above application is incorporated herein by reference.

FIELD

The present disclosure relates to a compressor with a shutdown assembly.

BACKGROUND

This section provides background information related to the present disclosure and is not necessarily prior art.

A climate-control system such as, for example, a heat-pump system, a refrigeration system, or an air conditioning system, may include a fluid circuit having an outdoor heat exchanger, an indoor heat exchanger, an expansion device disposed between the indoor and outdoor heat exchangers, and one or more compressors circulating a working fluid (e.g., a refrigerant) between the indoor and outdoor heat exchangers. Efficient and reliable operation of the one or more compressors is desirable to ensure that the climate-control system in which the one or more compressors are installed is capable of effectively and efficiently providing a cooling and/or heating effect on demand.

SUMMARY

This section provides a general summary of the disclosure and is not a comprehensive disclosure of its full scope or all of its features.

In one form, the present disclosure provides a compressor that may include first and second scrolls, and a shutdown assembly. The first scroll may be an orbiting scroll. The second scroll may be a non-orbiting scroll. The first and second scrolls have first and second spiral wraps, respectively, that meshingly engage each other to form a plurality of moving pockets. The second scroll includes a discharge passage that receives compressed working fluid from one of the pockets defined by the first and second spiral wraps. The shutdown assembly may include a valve housing and a valve member. The valve housing may be attached to the second scroll (i.e., the valve housing may be integrally formed with a part of the second scroll or the valve housing may be formed separately from the second scroll and mounted thereto). The valve housing may include a first aperture that receives working fluid from the discharge passage. The valve member may include a valve stem and a valve head. The valve stem may be movably received in a second aperture of the valve housing and has a smaller diameter than a diameter of the valve head. The valve member may be movable between a closed position in which the valve head restricts flow through the discharge passage and an open position in which the valve head allows flow through the discharge passage.

In some configurations of the compressor of the above paragraph, the valve head includes a first side that contacts a surface of the second scroll when the valve member is in the closed position, wherein the valve head includes a

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second side opposite the first side, and wherein a chamfer is formed on the second side at an outer periphery of the valve head.

In some configurations of the compressor of either of the above paragraphs, a raised boss is formed on the second side of the valve head, and wherein the raised boss is disposed radially between the valve stem and the chamfer.

In some configurations of the compressor of any one or more of the above paragraphs, the first side of the valve head includes a protrusion that extends into the discharge passage when the valve member is in the closed position.

In some configurations of the compressor of any one or more of the above paragraphs, the valve member includes a groove formed in an outer diametrical surface of the valve stem.

In some configurations of the compressor of any one or more of the above paragraphs, the groove is an annular groove that encircles a longitudinal axis of the valve stem.

In some configurations of the compressor of any one or more of the above paragraphs, the groove is an axially extending groove and extends in a direction parallel to a longitudinal axis of the valve stem.

In some configurations of the compressor of any one or more of the above paragraphs, the valve member includes at least another axially extending groove formed in the outer diametrical surface of the valve stem and extending in the direction parallel to the longitudinal axis of the valve stem.

In some configurations of the compressor of any one or more of the above paragraphs, a third aperture extends through at least a portion of the valve stem.

In some configurations of the compressor of any one or more of the above paragraphs, the third aperture extends through the valve head.

In another form, the present disclosure provides a compressor that includes an orbiting scroll, a non-orbiting scroll, and a discharge valve member. The non-orbiting scroll is meshingly engaged with the orbiting scroll and may include a lower scroll piece and an upper scroll piece that is mounted to the lower scroll piece. The upper scroll piece may include a central hub that defines a valve housing having a valve guide and a plurality of first apertures disposed around the valve guide. The discharge valve member may include a valve stem and a valve head disposed at an axial end of the valve stem. The valve head is disposed in a recess defined by a side of the upper scroll piece that faces the lower scroll piece. The valve stem may be movably received in a second aperture formed in the valve guide and has a smaller diameter than a diameter of the valve head. The valve member may be movable relative to the upper and lower scroll pieces between a closed position in which the valve head blocks a discharge passage in the lower scroll piece to restrict fluid flow therethrough and an open position in which the valve head allows fluid flow through the discharge passage.

In some configurations of the compressor of the above paragraph, the valve head includes a first side that contacts a surface of the lower scroll piece when the valve member is in the closed position, wherein the valve head includes a second side opposite the first side, and wherein a chamfer is formed on the second side at an outer periphery of the valve head.

In some configurations of the compressor of either of the above paragraphs, a raised boss is formed on the second side of the valve head, and wherein the raised boss is disposed radially between the valve stem and the chamfer.

In some configurations of the compressor of any one or more of the above paragraphs, the first side of the valve head

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includes a protrusion that extends into the discharge passage when the valve member is in the closed position.

In some configurations of the compressor of any one or more of the above paragraphs, the valve member includes a groove formed in an outer diametrical surface of the valve stem.

In some configurations of the compressor of any one or more of the above paragraphs, the groove is an annular groove that encircles a longitudinal axis of the valve stem.

In some configurations of the compressor of any one or more of the above paragraphs, the groove is an axially extending groove and extends in a direction parallel to a longitudinal axis of the valve stem.

In some configurations of the compressor of any one or more of the above paragraphs, the valve member includes at least another axially extending groove formed in the outer diametrical surface of the valve stem and extending in the direction parallel to the longitudinal axis of the valve stem.

In some configurations of the compressor of any one or more of the above paragraphs, a third aperture extends through at least a portion of the valve stem.

In some configurations of the compressor of any one or more of the above paragraphs, the third aperture extends through the valve head.

Further areas of applicability will become apparent from the description provided herein. The description and specific examples in this summary are intended for purposes of illustration only and are not intended to limit the scope of the present disclosure.

DRAWINGS

The drawings described herein are for illustrative purposes only of selected embodiments and not all possible implementations and are not intended to limit the scope of the present disclosure.

FIG. 1 is a cross-sectional view of a compressor having a shutdown assembly including a discharge valve member;

FIG. 2 is a cross-sectional view of a non-orbiting scroll and the shutdown assembly;

FIG. 3 is an exploded view of the non-orbiting scroll and the shutdown assembly of FIG. 2;

FIG. 4 is a partial cross-sectional view of a discharge valve member and upper and lower scroll pieces of a non-orbiting scroll with the discharge valve member in a closed position;

FIG. 5 is a partial cross-sectional view of the discharge valve member and upper and lower scroll pieces of FIG. 4 with the discharge valve member in an open position;

FIG. 6 is a partial cross-sectional view of another discharge valve member and the upper and lower scroll pieces of the non-orbiting scroll;

FIG. 7 is a perspective view of the discharge valve member of FIG. 6;

FIG. 8 is a partial cross-sectional view of yet another discharge valve member and the upper and lower scroll pieces of the non-orbiting scroll;

FIG. 9 is a perspective view of the discharge valve member of FIG. 8;

FIG. 10 is a partial cross-sectional view of yet another discharge valve member and the upper and lower scroll pieces of the non-orbiting scroll;

FIG. 11 is a partial cross-sectional view of yet another discharge valve member and the upper and lower scroll pieces of the non-orbiting scroll;

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FIG. 12 is a partial cross-sectional view of yet another discharge valve member and the upper and lower scroll pieces of the non-orbiting scroll; and

FIG. 13 is a perspective view of another discharge valve member.

Corresponding reference numerals indicate corresponding parts throughout the several views of the drawings.

DETAILED DESCRIPTION

Example embodiments will now be described more fully with reference to the accompanying drawings.

Example embodiments are provided so that this disclosure will be thorough and will fully convey the scope to those who are skilled in the art. Numerous specific details are set forth such as examples of specific components, devices, and methods, to provide a thorough understanding of embodiments of the present disclosure. It will be apparent to those skilled in the art that specific details need not be employed, that example embodiments may be embodied in many different forms and that neither should be construed to limit the scope of the disclosure. In some example embodiments, well-known processes, well-known device structures, and well-known technologies are not described in detail.

The terminology used herein is for the purpose of describing example embodiments only and is not intended to be limiting. As used herein, the singular forms “a,” “an,” and “the” may be intended to include the plural forms as well, unless the context clearly indicates otherwise. The terms “comprises,” “comprising,” “including,” and “having,” are inclusive and therefore specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof. The method steps, processes, and operations described herein are not to be construed as necessarily requiring their performance in the particular order discussed or illustrated, unless specifically identified as an order of performance. It is also to be understood that additional or alternative steps may be employed.

When an element or layer is referred to as being “on,” “engaged to,” “connected to,” or “coupled to” another element or layer, it may be directly on, engaged, connected or coupled to the other element or layer, or intervening elements or layers may be present. In contrast, when an element is referred to as being “directly on,” “directly engaged to,” “directly connected to,” or “directly coupled to” another element or layer, there may be no intervening elements or layers present. Other words used to describe the relationship between elements should be interpreted in a like fashion (e.g., “between” versus “directly between,” “adjacent” versus “directly adjacent,” etc.). As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items.

Although the terms first, second, third, etc. may be used herein to describe various elements, components, regions, layers and/or sections, these elements, components, regions, layers and/or sections should not be limited by these terms. These terms may be only used to distinguish one element, component, region, layer or section from another region, layer or section. Terms such as “first,” “second,” and other numerical terms when used herein do not imply a sequence or order unless clearly indicated by the context. Thus, a first element, component, region, layer or section discussed below could be termed a second element, component,

region, layer or section without departing from the teachings of the example embodiments.

Spatially relative terms, such as “inner,” “outer,” “beneath,” “below,” “lower,” “above,” “upper,” and the like, may be used herein for ease of description to describe one element or feature’s relationship to another element(s) or feature(s) as illustrated in the figures. Spatially relative terms may be intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. For example, if the device in the figures is turned over, elements described as “below” or “beneath” other elements or features would then be oriented “above” the other elements or features. Thus, the example term “below” can encompass both an orientation of above and below. The device may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein interpreted accordingly.

With reference to FIGS. 1-3, a compressor 10 is provided that may include a hermetic shell assembly 12, first and second bearing housing assemblies 14, 16, a motor assembly 18, a compression mechanism 20, a discharge port or fitting 24, a suction port or fitting 28, and a suction conduit (or suction funnel) 30.

As shown in FIG. 1, the shell assembly 12 may form a compressor housing and may include a cylindrical shell 32, an end cap 34 at an upper end thereof, a transversely extending partition 36, and a base 38 at a lower end thereof. The shell 32, the base 38 and the partition 36 may cooperate to define a suction-pressure chamber 39. The end cap 34 and the partition 36 may define a discharge chamber 40. The partition 36 may separate the discharge chamber 40 from the suction-pressure chamber 39. A discharge-pressure passage 43 may extend through the partition 36 to provide communication between the compression mechanism 20 and the discharge chamber 40. The suction fitting 28 may be attached to the shell assembly 12 at an opening 46.

As shown in FIG. 1, the first bearing housing assembly 14 may be disposed within the suction-pressure chamber and may be fixed relative to the shell 32. The first bearing housing assembly 14 may include a first main bearing housing 48 and a first bearing 50. The first main bearing housing 48 may house the first bearing 50 therein. The first main bearing housing 48 may fixedly engage the shell 32 and may axially support the compression mechanism 20.

As shown in FIG. 1, the motor assembly 18 may be disposed within the suction-pressure chamber 39 and may include a stator 60 and a rotor 62. The stator 60 may be press fit into the shell 32. The rotor 62 may be press fit on a drive shaft 64 and may transmit rotational power to the drive shaft 64. The drive shaft 64 may be rotatably supported by the first and second bearing housing assemblies 14, 16. The drive shaft 64 may include an eccentric crank pin 66 having a crank pin flat.

As shown in FIG. 1, the compression mechanism 20 may be disposed within the suction-pressure chamber 39 and may include an orbiting scroll 70 and a non-orbiting scroll 72. The first scroll member or orbiting scroll 70 may include an end plate 74 and a spiral wrap 76 extending therefrom. A cylindrical hub 80 may project downwardly from the end plate 74. A drive bushing may be disposed in the hub 80 and may receive the crank pin 66. An Oldham coupling 84 may be engaged with the orbiting and non-orbiting scrolls 70, 72 to prevent relative rotation therebetween.

As shown in FIG. 1, the second scroll member or non-orbiting scroll 72 may include an end plate 86 and a spiral wrap 88 projecting downwardly from the end plate 86. The spiral wrap 88 may meshingly engage the spiral wrap 76 of

the orbiting scroll 70, thereby creating a series of moving fluid pockets. The fluid pockets defined by the spiral wraps 76, 88 may decrease in volume as they move from a radially outer position (at a suction pressure) to a radially intermediate position (at an intermediate pressure) to a radially inner position (at a discharge pressure) throughout a compression cycle of the compression mechanism 20. As shown in FIGS. 1-3, a suction inlet 89 may be formed in the non-orbiting scroll 72 and may provide fluid communication between the suction conduit 30 and a radially outermost fluid pocket 93 (FIG. 1) formed by the spiral wraps 76, 88.

The non-orbiting scroll 72 may include an upper scroll piece 94 that is mounted to the end plate 86 (lower scroll piece). The upper scroll piece 94 is mounted to an upper surface 96 of the end plate 86. Fasteners (e.g., threaded fasteners) may extend through mounting apertures 98 (FIG. 3) of the upper scroll piece 94 and into mounting apertures 100 (FIG. 3) in the end plate 86. The upper scroll piece 94 may include an upper annular recess 102 that may receive a floating seal assembly 104, as shown in FIG. 1. The upper annular recess 102 may be defined by (and disposed radially between) a central hub 116 of the upper scroll piece 94 and an outer annular rim 157 of the upper scroll piece 94.

The upper scroll piece 94 may also include a lower recess 106 (FIG. 2) in which variable-compression-ratio valves 108 may be disposed. Each of the variable-compression-ratio valves 108 may include a valve member (e.g., a reed valve) 110 and a valve backer 112. The valve member 110 is movable relative to the end plate 86 to selectively open and close variable-compression-ratio ports 114 formed in the end plate 86. The variable-compression-ratio ports 114 are in fluid communication with intermediate-pressure pockets formed by the spiral wraps 76, 88. The variable-compression-ratio valves 108 selectively allow and prevent fluid communication between the intermediate-pressure pockets and the discharge chamber 40. The variable-compression-ratio valves 108 may be mounted to the end plate 86 by fasteners (e.g., pins or threaded fasteners) that engage apertures 109 (FIG. 3) in the end plate 86. A seal (e.g., O-ring) 107 may encircle the lower recess 106 and may sealingly engage the end plate 86 and upper scroll piece 94.

As shown in FIG. 2, the central hub 116 of the upper scroll piece 94 may define a valve guide (or valve housing) 118 and one or more apertures 120. The apertures 120 may be in fluid communication with the lower recess 106 and the discharge chamber 40. The valve guide 118 may movably engage (and guide movement of) a portion of a primary discharge valve member (or shutdown device) 122. For example, the valve guide 118 may include an aperture that reciprocatingly receives a stem 124 of the valve member 122. A head 126 of the valve member 122 may selectively open and close a discharge passage 128 formed in the end plate 86 to selectively allow and prevent fluid communication between the discharge passage 128 and the discharge chamber 40. The discharge passage 128 receives fluid from a discharge-pressure pocket formed by the spiral wraps 76, 88.

The suction conduit 30 may direct working fluid at a suction-pressure from the suction fitting 28 to the suction inlet 89 of the non-orbiting scroll 72 so that working fluid can be directed into the radially outermost fluid pocket 93 and subsequently compressed by the compression mechanism 20. As shown in FIGS. 1 and 2, a portion of the suction conduit 30 may snap into engagement with a wall 90 of the non-orbiting scroll 72 (e.g., a wall that defines a lower end of the suction inlet 89) and another portion of the suction

conduit **30** may be captured or clamped between the upper scroll piece **94** and the end plate **86**.

Referring now to FIGS. **4** and **5**, another upper scroll piece **194** (only partially shown) and another primary discharge valve member **222** are provided. The upper scroll piece **194** and valve member **222** may be incorporated into the compressor **10** instead of the upper scroll piece **94** and valve member **122**.

The upper scroll piece **194** may be similar or identical to the upper scroll piece **94** described above, apart from differences described below. Like the upper scroll piece **94**, the upper scroll piece **194** may include an outer annular rim (similar or identical to outer annular rim **157**) and a central hub **316** (similar to the central hub **116**) that cooperate to define an upper annular recess **202** (similar or identical to upper annular recess **102**). As shown in FIGS. **4** and **5**, the central hub **316** of the upper scroll piece **194** may be structured differently than the central hub **116**.

The central hub **316** may define a valve housing for the valve member **222**. In this manner, the central hub **316** and the valve member **22** cooperate to define a shutdown assembly (or discharge valve assembly) **302**. The central hub **316** may include a plurality of apertures **320** that extend therethrough. The apertures **320** provide fluid communication between a lower recess **206** (similar or identical to lower recess **106**) of the upper scroll piece **194** and the discharge chamber **40**. The central hub **316** may also include a valve guide **318** defining a guide recess **319**. The apertures **320** may be arranged in a circular pattern that encircles the guide recess **319**. Central aperture **321** may extend through a central portion of the central hub **316** and may be in fluid communication with the discharge chamber **40** and the guide recess **319**. A diameter of the central aperture **321** may be smaller than a diameter of the guide recess **319**,

The valve member **222** may include a stem **224** and a head **226**. The stem **224** may have a smaller diameter than the head **226**. The head **226** may be disposed at an axial end of the stem **224**. The stem **224** may be reciprocally received in the guide recess **319** such that the valve member **222** is movable relative to the upper scroll piece **194** and the end plate **86** between a closed position (FIG. **4**) and an open position (FIG. **5**).

In the closed position, the head **226** of the valve member **222** may contact the end plate **86** to prevent fluid flow through the discharge passage **128** (i.e., to prevent fluid communication between the discharge passage **128** and the discharge chamber **40**). In the open position, the head **226** of the valve member **222** is spaced apart from the end plate **86** to allow fluid flow through the discharge passage **128** and through the apertures **320** (i.e., to allow fluid communication between the discharge passage **128** and the discharge chamber **40** via the apertures **320**). When the compressor **10** shuts down, the valve member **222** is forced to move from the open position to the closed position. At shutdown of the compressor **10**, working fluid in the discharge chamber **40** may flow into the central aperture **321** and into the guide recess **319** to aid in forcing the valve member **222** toward the closed position.

As noted above, a diameter of the head **226** of the valve member **222** may be larger than a diameter of the stem **224** of the valve member **222**. In some configurations, a ratio of the diameter of the head **226** to the diameter of the stem **224** may range from 2-10.

The relatively large diameter of the head **226** allows the valve member **222** to cover (prevent fluid flow through) the discharge passage **128** during shutdown. The relatively small diameter of the stem **224** allows for: (a) large, drilled

discharge apertures **320** in the central hub **316** which reduce pressure drop through the shutdown assembly **302** and increase gas flow area and flow uniformity (which improves efficiency of the compressor **10**), (b) reduced mass of the valve member **222** (compared to single diameter shutdown devices) to reduce dynamic operation occurrence (which improves reliability of the shutdown assembly **302** and compressor **10**), and (c) reduced oil stiction of the valve member **222** to the valve guide **318** because of the small contact area along the guiding length (which improves shutdown response).

As shown in FIGS. **4** and **5**, the head **226** of the valve member **222** may include a chamfer **228** that faces away from the end plate **86** (i.e., the chamber **228** is located on a top side of the head **226** opposite a bottom side of the head **226** that contacts the end plate **86** in the closed position). The chamfer **228** allows for improved gas flow around the head **226** when the valve member **222** is in the open position (i.e., during operation of the compressor **10**), which improves efficiency of the compressor **10**.

Referring now to FIGS. **6** and **7**, another discharge valve member **422** is provided that may be incorporated into the compressor **10** (instead of the valve member **122**, **322**) with the upper scroll piece **194**. The valve member **422** may include some or all of the features of the valve member **222** described above (e.g., including a stem **424** and head **426** having some or all of the features of the stem **224** and head **226**). The stem **424** of the valve member **422** may also include a plurality of vertical grooves **430** (i.e., the grooves **430** extend in an axial direction parallel to a longitudinal axis of the stem **424**). The grooves **430** reduce the contact area between the stem **424** and the valve guide **318**, which reduces oil stiction of the valve member **422** to the valve guide **318** while still retaining sufficient alignment of the valve member **422** relative to the central hub **316** and end plate **86**.

Referring now to FIGS. **8** and **9**, another discharge valve member **522** is provided that may be incorporated into the compressor **10** (instead of the valve member **122**, **322**, **422**) with the upper scroll piece **194**. The valve member **522** may include some or all of the features of the valve member **222** described above (e.g., including a stem **524** and head **526** having some or all of the features of the stem **224** and head **226**). The stem **524** of the valve member **522** may also include one or more horizontal grooves **530** (i.e., the groove is an annular groove and extends in a circumferential direction and encircles the longitudinal axis of the stem **524**). The grooves **530** reduces the contact area between the stem **524** and the valve guide **318**, which reduces oil stiction of the valve member **522** to the valve guide **318** while still retaining sufficient alignment of the valve member **522** relative to the central hub **316** and end plate **86**.

Referring now to FIG. **10**, another discharge valve member **622** is provided that may be incorporated into the compressor **10** (instead of the valve member **122**, **322**, **422**, **522**) with the upper scroll piece **194**. The valve member **622** may include some or all of the features of the valve member **222**, **422**, **522** described above (e.g., including a stem **624** and head **626** having some or all of the features of the stem **224**, **424**, **524** and head **226**, **426**, **526**). The head **626** of the valve member **622** may also include a raised boss (or stepped-up portion) **630** formed on a top side of the head **626** (opposite a bottom side of the head **626** that contacts the end plate **86** in the closed position). The raised boss **630** has a smaller diameter than the outer diameter of the head **626** and the innermost diameter of chamfer **628** of the head **626**. The raised boss **630** may allow for reduced contact area with an

axial end surface 332 of the valve guide 318, which reduces oil stiction between the valve member 622 and the valve guide 318.

The head 626 of the valve member 622 may also include a protrusion 632 (e.g., an annular protrusion or disc-shaped protrusion) formed on the bottom side of the head 626 (e.g., on the side of the head 626 opposite the top side that includes the raised boss 630). The protrusion 632 may extend from the head 626 into the discharge passage 128. The protrusion 632 may aid in assembly of the shutdown assembly. For example, the protrusion 632 may act as a guide to position the valve member 622 on the end plate 86 prior to the upper scroll piece 194 being positioned on the end plate 86 and bolted thereto.

Referring now to FIG. 11, another discharge valve member 722 is provided that may be incorporated into the compressor 10 (instead of the valve member 122, 322, 422, 522, 622) with the upper scroll piece 194. The valve member 722 may include some or all of the features of the valve member 222, 422, 522, 622 described above (e.g., including a stem 724 and head 726 having some or all of the features of the stem 224, 424, 524, 624 and head 226, 426, 526, 626). The valve member 722 may also include an aperture 728 that may extend through some or all of the length of the stem 724 and/or may extend through some or all of the length of the head 726. The aperture 728 reduces the mass of the valve member 722, which may improve the efficiency of the compressor 10. Furthermore, the reduced thickness of the head 726 also reduces the overall mass of the valve member 722, which may improve the efficiency of the compressor 10.

Referring now to FIG. 12, another discharge valve member 822 is provided that may be incorporated into the compressor 10 (instead of the valve member 122, 322, 422, 522, 622, 722) with the upper scroll piece 194. The valve member 822 may include some or all of the features of the valve member 222, 422, 522, 622, 722 described above (e.g., including a stem 824 and head 826 having some or all of the features of the stem 224, 424, 524, 624, 724 and head 226, 426, 526, 626, 726). The valve member 822 may also include an aperture 828 that may extend through some or all of the length of the stem 824 and/or may extend through some or all of the length of the head 826. The valve member 822 may be formed from multiple pieces that may be welded or otherwise attached to each other. For example, the head 826 and the stem 824 may be formed separately from each other and then welded or otherwise attached to each other.

Referring now to FIG. 13, another discharge valve member 922 is provided that may be incorporated into the compressor 10 (instead of the valve member 122, 322, 422, 522, 622, 722, 822) with the upper scroll piece 194. The valve member 922 may include some or all of the features of the valve member 222, 422, 522, 622, 722, 822 described above (e.g., including a stem 924 and head 926 having some or all of the features of the stem 224, 424, 524, 624, 724, 824 and head 226, 426, 526, 626, 726, 826). The head 926 of the valve member 922 may include a chamfer 928 that faces away from the end plate 86 (i.e., the chamber 928 is located on a top side of the head 926 opposite a bottom side of the head 926 that contacts the end plate 86 in the closed position). The large chamfer 928 shown in FIG. 13 contacts the axial end surface 332 of the valve guide 318, which results in reduced contact area with the axial end surface 332, which reduces oil stiction.

The head 926 of the valve member 922 may also include a protrusion 932 (e.g., an annular protrusion or disc-shaped protrusion) formed on the bottom side of the head 926 (e.g.,

on the side of the head 926 opposite the top side that includes the chamfer 928). Like the protrusion 632 described above, the protrusion 932 may extend from the head 926 into the discharge passage 128 and may facilitate assembly.

Shutdown assemblies described above and/or shown in any of the figures have improved performance over the prior-art shutdown valves in several parameters including, for example, improved shutdown response time, reduced backflow leakage, reduced pressure loss across the shutdown assembly, reduced power loss, good stress and fatigue levels, good dynamic response, good impact velocity/force. Additionally, the shutdown assemblies according to the principles of the present disclosure provide the following advantages: reduced power loss compared to the prior-art shutdown valve by allowing more gas flow and/or more efficient gas flow; reduced noise by providing quicker and quieter compressor shutdowns by limiting backwards gas flow into the scroll set; reduced dynamic response; lower cost to manufacture & uses less parts than the prior-art shutdown valve; and easier to assemble and does not require adhesive.

The foregoing description of the embodiments has been provided for purposes of illustration and description. It is not intended to be exhaustive or to limit the disclosure. Individual elements or features of a particular embodiment are generally not limited to that particular embodiment, but, where applicable, are interchangeable and can be used in a selected embodiment, even if not specifically shown or described. The same may also be varied in many ways. Such variations are not to be regarded as a departure from the disclosure, and all such modifications are intended to be included within the scope of the disclosure.

What is claimed is:

1. A compressor comprising:

an orbiting scroll;

a non-orbiting scroll meshingly engaged with the orbiting scroll and including a lower scroll piece and an upper scroll piece that is fixedly mounted to the lower scroll piece, wherein the upper scroll piece includes a central hub that defines a valve housing having a valve guide and a plurality of first apertures disposed around the valve guide;

a discharge valve member having a valve stem and a valve head disposed at an axial end of the valve stem, wherein the valve head is disposed in a recess, wherein the recess is formed in a first surface of the upper scroll piece that faces the lower scroll piece, wherein a second surface of the lower scroll piece partially encloses the recess, and wherein the second surface of the lower scroll piece opposes the first surface of the upper scroll piece; and

variable-compression-ratio valves disposed within the recess and spaced apart from the discharge valve member, wherein the variable-compression-ratio valves selectively open and close respective variable-compression-ratio ports disposed in the lower scroll piece, wherein the valve stem is movably received in a second aperture formed in the valve guide and has a smaller diameter than a diameter of the valve head, wherein the valve member is movable relative to the upper and lower scroll pieces between a closed position in which the valve head blocks a discharge passage in the lower scroll piece to restrict fluid flow therethrough and an open position in which the valve head allows fluid flow through the discharge passage.

2. The compressor of claim 1, wherein the valve head includes a first side that contacts the second surface of the

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lower scroll piece when the valve member is in the closed position, wherein the valve head includes a second side opposite the first side, and wherein a chamfer is formed on the second side at an outer periphery of the valve head.

3. The compressor of claim 2, wherein a raised boss is formed on the second side of the valve head, and wherein the raised boss is disposed radially between the valve stem and the chamfer.

4. The compressor of claim 3, wherein the first side of the valve head includes a protrusion that extends into the discharge passage when the valve member is in the closed position.

5. The compressor of claim 1, wherein the valve member includes a groove formed in an outer diametrical surface of the valve stem.

6. The compressor of claim 5, wherein the groove is an annular groove that encircles a longitudinal axis of the valve stem.

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7. The compressor of claim 5, wherein the groove is an axially extending groove and extends in a direction parallel to a longitudinal axis of the valve stem.

8. The compressor of claim 7, wherein the valve member includes at least another axially extending groove formed in the outer diametrical surface of the valve stem and extending in the direction parallel to the longitudinal axis of the valve stem.

9. The compressor of claim 1, wherein a third aperture extends through at least a portion of the valve stem.

10. The compressor of claim 9, wherein the third aperture extends through the valve head.

11. The compressor of claim 1, wherein the variable-compression-ratio valves are mounted to the second surface of the lower scroll piece.

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